forces from one body to another, as if they were cables pulling and pushing (figure 2.4).

An object with an electric charge (a rubbed glass rod, for instance) distorts the electric and magnetic fields (the lines) around itself, and in turn these fields produce a force on each charged object immersed in them. Thus, two distant charged objects do not attract or repel each other *directly* but only via the medium interposed between them.

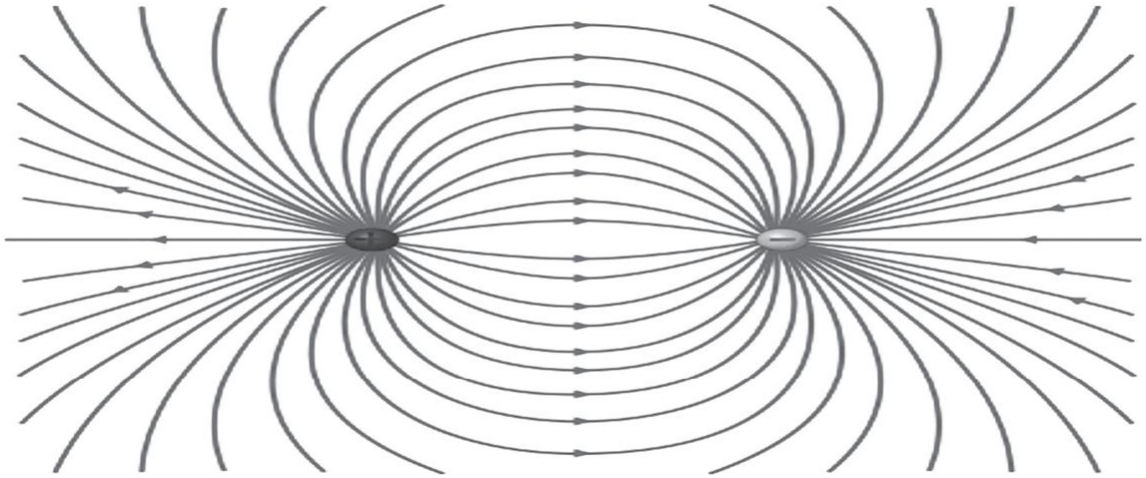


Figure 2.4 The field's lines fill space. Through them, two objects with electrical charge interact. The force of the two objects is 'carried' by the field's lines of force.

If you take two magnets in your hands and play with them, bringing them together and apart repeatedly, feeling the force with which they attract and repel, it is not difficult to experience the same intuition as Faraday, to 'feel' via these effects the *field* interposed between the magnets.

This is a radically different idea from the Newtonian notion of a force acting between distant bodies. But it would have appealed to Newton. Newton, indeed, was perplexed regarding this very attraction at a distance that he himself had introduced. How does the Earth manage to attract the Moon, which is so distant? How can the Sun attract the Earth without coming into contact with it? He had written in a letter:

> It is inconceivable that inanimate brute matter should, without the intervention of something else which is not material, operate upon and affect other matter, and have an effect upon it, without mutual contact.[5]

And, further down the page, we even find:

> That Gravity should be innate, inherent and essential to Matter, so that one Body may act upon another at a Distance thro' a Vacuum, without the Mediation of any thing else, by and through which their Action and Force may be conveyed from one to another, is to me so great an Absurdity, that I believe no Man who has in philosophical Matters a competent Faculty of thinking, can ever fall into it. Gravity must be caused by an Agent acting constantly according to